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ORIGINAL ARTICLE

The cenobamate KORK study—A prospective monocenter observational study investigating cenobamate as an adjunctive therapy in refractory epilepsy, with comparisons to historical cohorts treated with add-on lacosamide, perampanel, and brivaracetam

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Abstract

Objective: In Europe, cenobamate has been approved for use as an adjunctive therapy in adult patients with epilepsy (PWE) with focal-onset seizures (FOS) who have not responded satisfactorily to treatment with at least two antiseizure medications (ASMs). Pivotal trials and real-world observational studies have demonstrated a high efficacy of cenobamate, even in very difficult-to-treat epilepsies. Our aim was to investigate the efficacy of add-on cenobamate in adult PWE who were prospectively monitored. We compared these results with those previously obtained for add-on lacosamide, perampanel, and brivaracetam therapy.

Methods: Patients were enrolled from the CENKORK study, which is a prospective, non-interventional, open-label, monocenter cohort study of adult PWE experiencing FOS. The titration of cenobamate was performed according to the guidelines outlined in the summary of product characteristics. The primary outcome measure was the retention rate at 6 months and 1 year. In addition, we assessed seizure-free rates, the proportion of patients achieving at least a 50% seizure reduction, adverse events, and the reasons for treatment discontinuation. These outcome measures were compared with historical controls treated with adjunctive lacosamide, perampanel, or brivaracetam at our center.

Results: Between June 2021 and 2022, 172 PWE with ongoing FOS were included. 22 cases were lost to follow-up, leaving 150 cases for the 1-year assessment. The retention rates at 6 months and 1 year were 88.7% and 80%, respectively. Seizure freedom was achieved in 14% of patients at both the 6-month and 1-year marks, while the ≥50% responder rates were 50% and 61%, respectively. The 6-month retention rate was significantly higher in cenobamate than in other ASMs

Clinical trial registration: DRKS (DRKS-ID DRKS00030916).

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($p < 0.001$ for each comparator). Adverse events were significantly more common with perampanel ($p < 0.001$).

Significance: Add-on cenobamate proved to be particularly efficacious compared to our experience with other recently introduced ASMs.

Plain Language Summary: This observational study was carried out in 172 adult patients with difficult-to-treat epilepsy who were treated with adjunctive cenobamate. After 1 year, the data of 150 patients could be analyzed. Seizure freedom, in the preceding 3 months, was achieved in 14%. The rate of PWE continuing cenobamate was 80%. In our hands, cenobamate showed promising efficacy and tolerability even when compared to other recently introduced antiseizure medications.

KEYWORDS

cenobamate, drug-resistant, epilepsy, historical controls, monocenter, prospective

1 | INTRODUCTION

Epilepsy is one of the most prevalent chronic central nervous system disorders, affecting around 70 million people worldwide.[1] Despite the introduction of numerous new generation antiseizure medications (ASMs) over the last decade, recent findings from a meta-analysis of 39 studies spanning 1995–2021, involving over 20,000 patients with epilepsy (PWE), suggest that the rate of seizure freedom has not improved.[2] Thus, more effective ASMs with acceptable tolerability are urgently needed to address these persistently frustrating findings.

Cenobamate (CNB) is a novel ASM that was recently introduced to clinical practice. Its precise mode of action is yet to be fully elucidated.[2–7] So far, two potentially complimentary mechanisms have been delineated. The first involves the reduction of repetitive neuronal discharges by CNB via the inhibition of voltage-dependent sodium channels.[6–10] In particular, CNB augments the rapid and complete inactivation of sodium channels, and, akin to lacosamide (LCM), suppresses the non-activating component of the sodium current (I_{NaP}) in hippocampal rat neurons in a concentration-dependent manner.[6–13] The second potential mechanism of action involves γ-amino butyric acid (GABA).[6,7,10,12,14] CNB has previously been shown to potentiate GABA-induced currents in acutely isolated CA3 pyramidal cells in a concentration-dependent manner. Additionally, CNB acted as a positive allosteric modulator of high-affinity $GABA_A$ receptors, activated by GABA at a site independent of the benzodiazepine binding site, thereby enhancing tonic inhibition in hippocampal neurons.[8,10] This effect was similar for all tested $GABA_A$ receptors containing six different alpha subunits ($\alpha_1\beta_2\gamma_2$ or $\alpha_{2\text{-}6}\beta_3\gamma_2$). There is additional evidence that CNB interacts with both synaptic and extra-synaptic $GABA_A$ receptors, affecting both phasic (I_{phasic}) and tonic (I_{tonic}) $GABA_A$ currents.[10]

Key Points

- We performed a monocenter, non-interventional, prospective, observational study of add-on cenobamate in 172 adult patients with drug-resistant epilepsy.
- Cenobamate was compared with historical controls treated with add-on lacosamide, perampanel, or brivaracetam at our center.
- Data from 150 patients with epilepsy were analyzed.
- The retention rates at 6 and 12 months were 88.7% and 80%, respectively.
- The efficacy of cenobamate was higher than that of the historical controls.

In 2021, CNB was approved by the European Medicines Agency (EMA) for the adjunctive treatment of adult PWE with focal-onset seizures (FOS) who did not achieve a satisfactory response after appropriate treatment with at least two established ASMs. Its approval was based on the results of one proof-of-concept study,[11] two randomized placebo-controlled trials,[9,12] and one open-label safety study.[15]

In spite of the limitations of fixed-dose placebo-controlled trials, such as relatively short maintenance periods, several Phase-II trials have already suggested that CNB might be a particularly effective ASM, showing promise even when compared to the results of successful

similar studies with other recently investigated and approved ASMs.[16–20] Furthermore, commentaries have claimed that CNB could be a revolutionary drug in the field.[13,18,21] Indeed, in line with the pivotal trials, the open-label extension data and real-world studies supported this view.[17–20,22–28]

The Kork Epilepsy Center is one of the leading tertiary referral epilepsy centers in Germany.[29] We had the privilege of participating in two pivotal trials with CNB,[12,15] providing us with firsthand experience in the practical administration and the potential benefits that CNB offers.

The CENKORK study describes the experiences of a cohort with PWE that we prospectively followed from the introduction of CNB in Germany onwards. This initiative commenced immediately after the conclusion of a compassionate use program. We initiated this study independently to avoid that these data might be included in additional studies conducted by the manufacturer or others.

Since many of our patients suffer from long-lasting and difficult-to-treat epilepsies, it is self-explanatory that we typically have a substantial number of PWE awaiting a new treatment option. Therefore, we were able to compare our previously published findings from our single-center experiences with new generation ASMs, including LCM, perampanel (PER), and brivaracetam (BRV)[30–32] with our results obtained with CNB and presented in this paper.

2 | METHODS

We prospectively followed in- and out-PWE who were treated with adjunctive CNB due to their difficult-to-treat epilepsies. CENKORK is an ongoing non-interventional study conducted in accordance with the Declaration of Helsinki code of ethics and following the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines.[33] CENKORK was approved by the local ethics committee of the Medical Faculty at the University of Freiburg, Germany (No. 22-1139) and is registered with the German registry of clinical studies DRKS (DRKS-ID DRKS00030916). Informed consent was provided by each participant.

Since the primary objective of the study was to gather real-world experience, the only inclusion criteria were adulthood and the decision to initiate add-on CNB due to ongoing seizures despite prior and ongoing ASM treatment. The reason to initiate CNB was not influenced by participation in the study. To ensure an appropriate assessment of efficacy, PWE were included only if they reported a minimum seizure frequency of one or more unequivocal focal seizures per month during the preceding 3 months, based on their self-reports and seizure calendars. The exclusion criteria were age under 18 years, PWE with legal representatives, idiopathic generalized epilepsy syndromes, insufficient data reporting, and individuals with previously diagnosed psychogenic non-epileptic seizures.

We assessed various demographic data including age, gender, age at epilepsy onset, seizure and epilepsy classification, seizure etiology, previous ASMs and the number of previous ASMs administered, current ASM treatment, and the seizure frequency during the 3 months prior to the initiation of CNB. Data were de-identified prior to the data analysis.

During the prospective 1-year period of CNB titration and dosing, evaluations were conducted to assess seizure frequency and severity, as well as adverse events (AEs). PWE were followed on an individual basis. For every PWE included in the final analysis, direct communication was maintained, enabling dose adjustments of CNB and concomitant drugs to be made at any time, extending beyond the regular 3-month physical or telephone visits. Prior to the introduction of CNB to the market, we were able to establish our own therapeutic drug monitoring (TDM) assay in 2021. In the majority of PWE, we were able to assess the plasma levels of CNB. The results of these investigations and a discussion regarding a potential therapeutic range are addressed in another paper. Here, our focus is on presenting data that were continuously available at every time point throughout the prospective 1-year investigation.

The primary outcome variable was the retention rate after 6 and 12 months. PWE lost to follow-up were not considered for the final analysis. Additional outcome measures assessed during CNB treatment included the dosage of CNB and the concomitant medications, seizure frequency and severity, and AEs. AEs were documented by means of physical or telephone visits at 3, 6, 9, and 12 months after the initiation of CNB treatment. When AEs occurred earlier, the PWE contacted us to let us know. Emergency visits were always possible. In instances of a premature discontinuation of CNB, the PWE would be counted as a drop out at the subsequent regular timepoint. Thus, an AE leading to the discontinuation of CNB 6 weeks after initiation would have been included in the 3-month assessment and so on. The evaluation of seizure frequency also comprised the 3 months preceding each timepoint, comparing it with the 3 months prior to the initiation of add-on CNB for comparison purposes. CNB was titrated according to the schedule used in the Phase-III trial conducted by Sperling et al.,[15] which is also recommended in the summary of product characteristics (SmPC) for the European Union. Thus, the intended dosing regimen involved once-daily administration, starting with 12.5 mg of CNB during the first 2 weeks of titration, followed by dose increases to 25 mg at weeks three and four, 50 mg at weeks five and six, and 100 mg thereafter. At this timepoint, we regularly contacted the participants to communicate the

further strategy according to the efficacy and tolerability at Week 7. In every case, we aimed to optimize dosing on an individual basis, with 400 mg as the maximum dose, following the results of the pivotal trial by Krauss et al.[12] and the European SmPC. In single cases, we did not rule out higher dosing if efficacy and tolerability permitted.

Additional analyses involved examining the results associated with CNB outcomes combined with strong enzyme inducers, which included carbamazepine (CBZ), oxcarbazepine (OXC), eslicarbazepine acetate (ESL), phenytoin (PHT), and phenobarbital (PB), as well as non-enzyme inducers (all other ASMs). Further analyses investigated the effect of concomitant sodium channel blockers (CBZ, OXC, ESL, PHT, lamotrigine [LTG], LCM) on the efficacy of CNB.

The second part of this study was the comparison of the present results with our previously published experiences of other recently labeled ASMs, namely adjunctive LCM, PER, and BRV.[30,32] Six-month retention rates, ≥50% responder rate, seizure freedom rates, and AEs resulting from add-on CNB were compared with the results obtained with add-on LCM ($n = 70$), PER ($n = 70$), and BRV ($n = 101$).

2.1 | Statistics

For the prospective observational CENKORK study, we used descriptive statistics to characterize the demographic features of the study population. All data were collectively analyzed and stratified by treatment populations. Numeric data, such as age, are described by the following statistical parameters: valid N, missing N, mean, minimum, mean, median, and maximum values. Categorical data, such as sex, are presented by means of *n* (%). The calculation of percentages was based on the number of patients with valid data for each parameter. The endpoints included retention rates, seizure-free occurrences, and ≥50% responder rates for the preceding 3 months, with the latter compared with the three-month baseline. For subgroup analyses of the demographic characteristics, only side-by-side comparisons with no statistical analysis were performed. No controls for confounding effects were performed. Since this was a non-interventional study, no formal sample size calculations were performed.

For the comparison with our single-center studies on add-on LCM, PER, and BRV, we conducted a retrospective, single cohort, comparative study to assess the efficacy and tolerability of CNB as add-on therapy for PWE with drug-resistant focal epilepsies. Here, we compared the 6-month retention rate, the percentage of PWE who were seizure-free, the percentage of responders achieving at least a 50% reduction in seizures, and the rate of AEs after an observation period of 6 months. We chose the results after 6 months because our studies on the other ASMs were also conducted within this timeframe. All PWE were adults with drug-resistant focal epilepsies who had been treated with CNB, LCM, PER, or BRV for at least 6 months. Data were collected from three studies conducted between 2016 and 2023, including the prospective CENKORK study, which forms a major part of this manuscript. The retention rate was defined as the proportion of patients who continued to take the medication at 6 months. The ≥50% responder rate was defined as the proportion of patients who had a 50% or greater reduction in seizure frequency at 6 months. The seizure-free rate was defined as the proportion of patients who had no seizures during the preceding 3 months at the 6-month follow-up. AEs were recorded using the National Institutes of Health Common Terminology Criteria for Adverse Events (CTCAE) version 5.0.

Categorical variables were summarized as proportions. Chi-squared tests were used to compare retention rates, ≥50% responder rates, and seizure-free rates between CNB and the other three ASMs. Since we conducted multiple comparisons (one for each of the four medications), the Bonferroni correction was applied to adjust the significance level. Bonferroni correction reduces the significance level by dividing it by the number of comparisons. In this case, the adjusted significance level was 0.017. The threshold for statistical significance was set at $p < 0.05$. R software version 4.3.2 (http://www.r-project.org) was used to perform the statistical analysis.

3 | RESULTS

Between June 2021 and June 2022, when CNB was marketed in Germany, we prospectively and consecutively recruited 172 adult PWE for whom treatment with add-on CNB was initiated. The patients had a mean age of 40.5 ± 14.4 years (median 39 years, range 18–76 years) and 91 were females (52.9%). The average duration of epilepsy was 26.6 ± 15 years (median 25 years, range 1–61 years). The mean age at the onset of epilepsy was 13.9 ± 14.3 years (median 10 years, range 0–70). All patients had FOS at baseline. Focal to bilateral tonic–clonic seizures were reported in 108 PWE (62.8%). The etiology was structural in 67% of patients with magnetic resonance imaging-detected lesions. The number of ASMs prior to the use of CNB varied between 2 and 27 (mean 10.2 ± 4.2, median 10). When CNB was initiated the mean number of baseline ASMs was 2.6 ± 1 (median 3, range 1–6), with 77.9% of patients undergoing treatment with two or three ASMs. 25 various ASMs were represented among the concomitant ASMs at the initiation of CNB. The most frequently used ASMs included

LTG, BRV, LCM, LEV, and valproic acid (VPA), the latter was typically combined with LTG. Seven patients had undergone epilepsy surgery. Of these, two were treated with vagus nerve stimulation and deep-brain stimulation. Three of the remaining five patients (59.9%) had daily or weekly seizures despite the baseline medication. At the time of recruitment, 38 were in-patients and 134 were out-patients. The last patient was included in June of 2022. The data lock for this report occurred in June 2023.

During the year after the initiation of CNB, 22 PWE were lost to follow-up. All of them were in-patients who did not seek our continued advice and supervision. Thus, 150 PWE were included in the final analysis. Demographic data of the included patients are summarized in Table 1.

TABLE 1 Demographic data.

Total number of PWE	172
Female, *n* (%)	91 (52.9%)
Number of in-patients	38
Number of out-patients	134
Number of PWE lost to follow-up	22
Mean age, years (SD; range)	40.5 (14.4; 18–76)
Median age, years	39
Mean time since diagnosis, years (SD; range)	26.6 (15; 1–61)
Median time since diagnosis, years	25
Mean age at onset of epilepsy, years (SD; range)	13.9 (14.3; 0–70)
Median age at onset of epilepsy, years	10
Focal epilepsy	171 (99.4%)
Combined focal and generalized epilepsy	1 (0.6%)
Structural etiology	115 (66.8%)
Genetic etiology	3 (1.7%)
Non-lesional/unclear	54 (31.4%)
Focal aware seizures	100 (58.1%)
Focal unaware seizures	150 87.2%)
Focal to bilateral tonic–clonic seizures, *n* (%)	108 (62.8%)
Daily seizures	61 (35.5%)
More than one seizure per week	42 (24.4%)
More than one seizure per month and less than one seizure per week	69 (40.1%)
Number of baseline ASMs prior to cenobamate	
Mean (SD; Range)	10.2 (4.2; 2–27)
Median	10
Number of baseline ASMs on cenobamate initiation	
Range	1–6
Mean	2.6 ± 1
Median	3
Patients with 2–3 ASMs	134 (77.9%)

Abbreviations: ASM, antiseizure medication; PWE, patients with epilepsy; SD, standard deviation.

3.1 | Dosing

After 3 months, the mean daily dose of CNB was 175.2 mg (range, 12.5–400 mg). After 6 months, the mean daily dose of CNB was 238.9 mg (range, 50–500 mg), and after 12 months the mean daily dose was 261.9 mg (range, 50–500 mg).

3.2 | Retention rate

Among the 150 included PWE, the 6-month retention rate was 88.7% ($n = 133$). The 12-month retention rate was 80% ($n = 120$). The retention rates at 3 and 9 months were 94% ($n = 141$) and 84.7% ($n = 127$), respectively.

Thus, CNB was discontinued in a total of 30 PWE (20%). Reasons for discontinuation in these PWE included lack of efficacy ($n = 16$), AEs ($n = 12$) and were unclear in a further two PWE. Rash ($n = 2$), somnolence ($n = 1$), dizziness ($n = 1$), and headaches ($n = 1$) were the predominant AEs reported within the first 3 months; somnolence ($n = 1$), nervousness ($n = 1$), and gynecomastia ($n = 1$) were the primary AEs occurring after 6 months; and gynecomastia ($n = 1$), increased liver parameters ($n = 1$), headaches ($n = 1$), and ataxia ($n = 1$) were the AEs most frequently reported after 1 year. Both PWE with rashes were re-exposed to CNB and had no relapse. Thus, we assumed that the rashes were not related to CNB. In addition, there were no further signs of any systemic systems or abnormal laboratory findings.

3.3 | Efficacy

Complete freedom of seizures during the preceding 3 months was achieved in 14% of PWE after 6 months ($n = 19$) and 1 year ($n = 17$). The ≥50% responder rates were 50% ($n = 69$) and 61% ($n = 75$) after 6 months and 1 year, respectively. The mean daily CNB dose in PWE who were seizure-free after 6 months was 218 mg (range, 100–400 mg). After 1 year, the daily doses ranged between 100 mg and 400 mg (mean dose, 208 mg).

At baseline, 98 PWE had focal to bilateral tonic–clonic seizures (FBTCS). These were abolished in 24 patients after 6 months (24%), and in 21 after 1 year (21%). Daily CNB doses ranged between 100 mg and 400 mg (mean, 279 mg) for FBTCS-free PWE after 12 months.

In six PWE who were seizure-free after 6 months, seizure relapse occurred between Months 6 and 12 (31.5%)

at daily doses ranging between 150 mg and 400 mg. In all of these patients, seizure relapses occurred following the reduction or discontinuation of concomitant ASMs (LTG, LEV, and LCM). In four of them, dose increases led to further seizure freedom thereafter with a minimum observation period of 6 months. The remaining two PWE were satisfied with their seizure situation and did not request a further increase in their dosage.

4 | ADVERSE EVENTS

After 3 months, 28 PWE complained of AEs (18.7%). AEs were the primary reason for discontinuation after 3 months of CNB therapy in all nine PWE. The leading AE was somnolence (46.4%), followed by dizziness (32%) as well as ataxia and nervousness (10.7% each). If several AEs occurred, combinations of somnolence, ataxia, dizziness, nystagmus, or diplopia were observed to varying degrees. We conducted a detailed examination of the concomitant medications used by these PWE. All of them, except for one patient, were on clobazam (CLB) or sodium channel blockers such as LCM, LTG, OXC, or ESL. The incidence of AEs in PWE taking sodium channel blockers was 17.4% and 8.7% in those without sodium channel blockers. Due to the small number of PWE without sodium channel blockers, we did not perform a statistical comparison. Still, among the PWE who did not experience AEs, two received unusually high daily doses of LCM, specifically 800 mg and 1100 mg, respectively.

PWE who were administered the concomitant enzyme inhibitor VPA had more AEs (34.3% of the follow-ups, $n = 12$ of 35) than PWE without enzyme-inhibiting ASMs (15% of the follow-ups, $n = 98$ of 653). The frequency of AEs was identical in PWE with and without enzyme-inducing ASMs (43.2% each: 16 PWE out of 37 with enzyme-inducing drugs in at least 1one follow-up, 63 PWE out of 146 PWE without enzyme-inducing drugs in at least one follow-up).

After 6 months, AEs were apparent in 43 of the remaining 133 PWE (32.3%). Three PWE discontinued CNB for tolerability reasons. Somnolence was by far the most common AE (60.4%, $n = 26$), followed by ataxia (16.3%, $n = 7$), and dizziness (11.6%, $n = 5$).

After 12 months, a further four PWE decided to discontinue CNB for tolerability reasons. AEs were reported in 32.5% ($n = 39$) of the patients who continued CNB after 12 months ($n = 120$). Again, somnolence was the most common AE (41%, $n = 16$), followed by dizziness (20.5%, $n = 8$) and ataxia (12.8%, $n = 5$). Altogether, AEs occurred in 74 of 150 PWE during the 12-month observation period (49.3%).

In two male participants, gynecomastia occurred without any laboratory or endocrinological explanations and was completely reversible after the discontinuation of CNB. Both these patients refused to be re-exposed to CNB. The results are displayed and summarized in Table 2.

The complete list of AEs is shown in Table S1.

4.1 | Changes in the concomitant medication

In 124 PWE (82.7%), concomitant ASMs were changed during the treatment with CNB. The number of concomitant ASMs was reduced in 59 cases (39.3%). More than one ASM (range, 2–3) were discontinued in 13 PWE (8.7%).

The number of concomitant ASMs increased in four cases (2.9%) and remained unchanged in 61 cases (40.7%) due to an exchange of the concomitant ASMs. Details of these changes are shown in Table 3.

TABLE 2 Results.

Mean daily dose after 6 months	238.9 mg (50–500 mg)
Mean daily dose after 12 months	261.9 mg (50–500 mg)
Retention rate after 3 months	94% ($n = 141$)
Retention rate after 6 months	88.7% ($n = 133$)
Retention rate after 9 months	84.7% ($n = 127$)
Retention rate after 12 months	80% ($n = 120$)
Discontinuation rate after 12 months	20% ($n = 30$)
Reason for discontinuation	
Lack of efficacy	10.7% ($n = 16$)
Adverse events	8% ($n = 12$)
Unclear	1% ($n = 2$)
≥50% responder rate after 6 months	50% ($n = 69$)
≥50% responder rate after 12 months	61% ($n = 75$)
Seizure-free PWE after 6 months	14% ($n = 19$)
Seizure-free PWE after 12 months	14% ($n = 17$)
Seizure freedom from focal to bilateral tonic–clonic seizures after 6 months	24% ($n = 24$)
Seizure freedom from focal to bilateral tonic–clonic seizures after 12 months	21% ($n = 21$)
Adverse events after 3 months	18.7% ($n = 28$)
Adverse events after 6 months	32.3% ($n = 43$)
Adverse events after 12 months	32.5% ($n = 39$)

Note: ≥50% responders = PWE with a reduction in seizure frequency by at least 50% during the preceding 3 months compared with the baseline prior to the initiation of cenobamate. Seizure-free PWE = Completely seizure-free PWE during the preceding 3 months compared with the baseline prior to the initiation of cenobamate.

Abbreviation: PWE, patients with epilepsy.

TABLE 3 Patients with epilepsy with changes of concomitant antiseizure medication during the treatment with cenobamate.

		N	%
Total number of PWE		150	100
PWE with changes of concomitant ASMs during treatment with cenobamate		124	82.7
PWE without changes of concomitant ASMs		26	17.3
Stable number but exchange of concomitant ASMs		61	40.7
Number of concomitant ASMs prior to the initiation of cenobamate	Number of concomitant ASMs with cenobamate (last observation carried forward)		
1	1	10	6.7
2	2	35	23.4
3	3	11	7.3
4	4	5	3.4
Differing number of concomitant ASMs during treatment with cenobamate		63	42
Reduced number of concomitant ASMs		59	39.3
Number of concomitant ASMs prior to the initiation of cenobamate	Number of concomitant ASMs with cenobamate (last observation carried forward)		
2	1	17	11.6
3	2	25	16.7
4	3	3	2.1
5	4	1	0.7
3	1	9	6.0
4	2	2	1.4
5	3	1	0.7
5	2	1	0.7
Increased number of concomitant ASMs		4	2.7
Number of concomitant ASMs prior to the initiation of cenobamate	Number of concomitant ASMs with cenobamate (last observation carried forward)		
2	3	2	1.4
3	4	2	1.4

Abbreviation: ASM, antiseizure medication.

4.2 | Comparison with LCM, PER, and BRV

Table 4 presents comparisons of the current findings with our previously published experiences regarding add-on LCM, PER, and BRV as adjunctive treatments.[30,32] Demographic data of these historical cohorts were similar to those of the CNB group in the present study. The etiology was structural in 70% of cases undergoing treatment with LCM and PER, and in 71% of cases treated with BRV. The mean ages of patients treated with LCM, PER, and BRV were 43 years, 39 years, and 42 years, respectively. The median previous number of ASMs were 7, 9, and 10, respectively. The only notable difference concerning the number of concomitant ASMs was observed between LCM and BRV, with 43% and 35%, respectively, being on monotherapy, while 56% and 60% were on two or three ASMs. In contrast, with PER, most PWE had been on two or three ASMs (76%), a pattern similarly observed with CNB (78%).

TABLE 4 Experiences with add-on lacosamide, perampanel, brivaracetam, and cenobamate 6 months after the initiation at the Kork Epilepsy Center.

	Lacosamide *N* = 70	Perampanel *N* = 70	Brivaracetam *N* = 101	Cenobamate *N* = 150
Retention rate	65%	67%	52%	89%*
≥50% responder	29%	49%	28%	50%
Seizure-free	4%	14%	7%	14%
Adverse events	46%	73%**	37%	42%

Note: Comparison of the actual results with historical cohorts treated with adjunct lacosamide, perampanel, or brivaracetam according to previously published monocenter studies at the Kork Epilepsy Center.[30,32] ≥50% responders = PWE with a reduction in seizure frequency by at least 50% during the preceding 3 months compared to the baseline prior to the initiation of lacosamide, perampanel, brivaracetam, or cenobamate. Seizure-free PWE, completely seizure-free PWE during the preceding 3 months compared to the baseline prior to the initiation of lacosamide, perampanel, brivaracetam, or cenobamate.

*$p < 0.001$ cenobamate versus lacosamide, perampanel, and brivaracetam. **$p < 0.001$ cenobamate versus perampanel all other differences statistically not significant.

The retention rates differed significantly ($p < 0.001$) in favor of CNB. AEs were significantly less frequent ($p < 0.001$) with CNB compared to PER. No significant differences were found between CNB and the other three ASMs concerning the rate of ≥50% responders and seizure freedom (see Table 4).

5 | DISCUSSION

To the best of our best knowledge, this is the second independent prospective observational real-world study to investigate the efficacy of add-on CNB. Novitskaya et al. performed a study with an almost identical design and achieved similar results.[28] Other previous prospective surveys were open-label follow-up studies of pivotal randomized controlled trials[17,22,23] or the open-label Phase-III trial that addressed safety as the primary outcome criterion.[15]

The mean daily dose of CNB of 175.2 mg after 3 months reflected that some PWE were still in the up-titrating process. This mean dose was almost identical to that reported by Villanueva et al.[19] After 6 and 12 months, the mean daily doses were in the range of 250 mg, as in other publications.[15,19,23]

In line with previous results of the pivotal and post-marketing studies,[9,12,15,17,19,20,22–28] our data confirm that CNB is a highly efficacious ASM.

The characteristics of PWE treated with CNB in previously published real-world studies were very similar to our PWE group, representing a population with very difficult-to-treat epilepsies. First, the mean duration of epilepsy prior to CNB spanned decades, and previously applied ASMs were numerous in our PWE group as in the cohorts of other studies.[24,26–28] Second, the previous number of ASMs was 10.5 in our study and 8–12 in other reports.[24,26–28] Finally, consistent with findings from other studies, most of the PWE investigated in our study were on two to three concomitant ASMs,[15,28] and even on three or more ASMs in other publications.[20,26]

5.1 | Retention rates

The primary outcome variables were the six-month and 12-month retention rates. Given the highly refractory nature of the cohort of PWE in this study, many of whom had endured drug-resistant courses for decades, with a median number of previously applied ASMs of 10 and the majority of PWE being treated with two or three concomitant ASMs at the initiation of CNB, the 12-month retention rate of 80%, along with the 6-month retention rate of almost 89%, reflects promising efficacy. Retention rates serve as the best real-world measure of achieving a satisfying balance of long-term efficacy, safety, tolerability, and adherence.[17] This again confirms the results of previous reports. The pooled data of 1844 PWE who entered open-label follow-up studies of the double-blind placebo-controlled studies had a 1-year retention rate of 80%.[17] Other studies reported 1-year retention rates of 87%,[26] 83%,[23] and 79%.[15]

More PWE discontinued CNB due to a lack of efficacy, suggesting that AEs, which were apparent in 49.3% of the whole group at least at one timepoint of the 1-year observation period, could typically be managed by modifying the drug load accordingly to overcome AEs, thus allowing the treatment to be continued. Several expert recommendations proposed how to deal with such tolerability issues, especially considering pharmacokinetic and pharmacodynamic interactions with concomitant ASMs such as CLB, LTG, PHT, PB, OXC, or LCM.[34–38] Our study was initiated immediately after the introduction of CNB to the free market following the conclusion of the compassionate use program offered by the manufacturer in Germany.

It is worth noting that we did not adhere to certain recommendations, such as proactive reductions of baseline ASMs under certain clinical conditions.[37] We suppose that in some instances such strategies might have been helpful to further reduce AEs and the discontinuation rates.

Most PWE discontinued CNB because of a lack of efficacy, like in the study by Klein et al.[23] Others reported that there were more discontinuations due to AEs than due to a lack of efficacy.[17,22] The most significant finding across all studies is the high retention rates observed, indicating that neither lack of efficacy nor AEs were so severe that they would have a major impact on the retention of CNB.

There may have been interfering factors that led to a false-positive high retention rate. CNB retention rates were potentially influenced by the fact that PWE may have considered to stay on CNB as it the most recently introduced ASM to the market. Additionally, there may have been a lack of alternative treatment options available, leading patients to remain on CNB even if their treatment response was questionable or less satisfying.

5.2 | Efficacy

Efficacy was assessed after 6 and 12 months. We collected data of the clinical course from the first day of CNB treatment onwards but decided not to investigate the efficacy of CNB at 3 months for methodological reasons.

The guidelines recommended a gradual titration of CNB due to reported cases of drug reaction with eosinophilia and systemic symptoms syndrome during the early development program. This evidence underscores the importance of avoiding such reactions through more cautious dosing strategies.[15] Therefore, the steady state of recommended daily maintenance doses, spanning 100 mg to 400 mg, is usually attained after several weeks to months.[3,5–7,39–41] Still, early beneficial effects on seizure frequencies at rather low daily doses were reported earlier both by post hoc analyses of the pivotal trials and from observational real-world studies.[24,28,42,43] We still did not analyze efficacy data after 3 months in detail. In many cases, the expected maintenance dose had not yet been established recommended slow titration schedule. Consequently, we assumed that it might be too early to utilize the first 3 months for efficacy analysis as has been done by others.[24,28] Furthermore, considering that the plasma steady state is typically reached only after approximately 14 days following a change in dosing and that even a maintenance dose of 100 mg is established only after 8 weeks according to the recommended dosing schedule, it would have been questionable to compare this period with the preceding 3 months prior to the initiation of CNB in terms of efficacy. Therefore, it is rather difficult to explain why in patients who were seizure-free immediately after the initiation of CNB the spontaneous course should not have been a confounding factor. In other words, we had reservations about asserting that patients became seizure-free within the first 3 months of treatment were unequivocal responders to CNB, considering that it takes several weeks to achieve a daily dose of 100 mg. Therefore, during the titration period of CNB we focused primarily on tolerability.

At both the 6- and 12-month marks, 14% of PWE were seizure-free. 50% and 61% of PWE were ≥50% responders at both timepoints, respectively. Again, considering the PWE cohort of hitherto were completely drug-resistant and had highly active epilepsies, these results remarkably emphasize the efficacy of CNB.

Generalized tonic–clonic seizures are widely recognized as the most devastating and dangerous seizure type.[44,45] With CNB, one in four of our PWE became seizure-free after 6 months, and one in five were seizure-free after 1 year. Even higher rates were reported by other studies, namely 55% according to Peña-Ceballos et al.,[27] albeit after a shorter observation period.

Our results are very similar to other real-world experiences from various countries and centers, regardless of whether the study was monocentric or multicentric.[15,17,19,20,22–27] Similarly, in a prospective study by Novitskaya et al.,[28] seizure freedom was reported in 5% of patients after 6 months and in 19% after 1 year. Furthermore, in their study the ≥50% responder rates were 40% and 55%, respectively.

During the first 3 months, that is, during the titration period, we observed a trend toward worsened tolerability when CNB was added to VPA and sodium channel blockers. Conversely, AEs were equally frequent when CNB was combined with enzyme-inducing ASMs or with ASMs without enzyme-inducing properties. Since VPA does not exert an impact on the plasma levels of CNB[5], we suggest that the higher rate of AEs was due to CNB itself, whereas the higher rate of AEs with sodium channel blockers resulted from the well-known pharmacodynamic interaction that has already been described and reported.[34–38] Neither enzyme-inducing ASMs nor sodium channel blockers had an impact on the outcomes in a similar study by Novitskaya et al.[28] Thus, these potential confounding factors should be kept in mind in the future but should not be overemphasized.

The daily CNB doses among seizure-free PWE after 12 months ranged between 100 mg and 400 mg (mean, 215 mg). The mean doses did not differ between responders and non-responders. In fact, they tended to be higher in the latter group. In contrast to other studies,[26,27,43] but in line with a recent publication,[28] some PWE even experienced significant benefits from lower doses of CNB doses,

to the extent that they became seizure-free. It is tempting to speculate that these PWE were particularly sensitive to the novel GABAergic mode of action. Similarly, we aim to discuss that the loss of efficacy, characterized by seizure relapses after initial freedom of seizures observed in seven PWE, as described by Peña-Ceballos et al.,[27] might also result from this mode of action. This is because a loss of efficacy over time is well-known phenomenon associated with other ASMs with GABAergic properties.[46]

5.3 | Tolerability and AEs

In contrast to efficacy, tolerability was assessed during the titration period to better understand what kind of AEs typically occur and whether the recommended schedule is sufficient to avoid too many AEs from occurring that might hamper the adherence of PWE who are already frustrated after the disappointing use of 10 or more ASMs in the past. We did observe a higher incidence of AEs during the first 3 months of treatment, suggesting that the recommended slow titration strategy likely reduces the risk of experiencing significant and numerous AEs during the first weeks of treatment with CNB.

Independent of the timepoint of evaluation, the leading AEs were somnolence, dizziness, and ataxia which occurred more often in combination with sodium channel blockers. This finding supports the recommendations to consider early reduction of sodium channel blockers with sedating profiles during titration.[34–38] On the other hand, CNB was well tolerated among most PWE despite unfavorable prerequisites. In our study, proactive modifications of the drug regimen were not necessary. We recommend proactive and comprehensive education of patients with epilepsy (PWE) and their relatives regarding potential AEs, coupled with well-organized communication. With such information provided, individuals should still be able to react promptly and appropriately if AEs occur.

Rare idiosyncratic AEs were a major drawback of several ASMs in the past. We do not want to overemphasize the finding of two cases of gynecomastia in male PWE. However, the community should be aware of this observation and closely follow and report similar cases.

Because of the recommendations given by the European Medicine Agency, and in contrast to those of the Food and Drug Administration in the United States, CNB is not licensed for monotherapy in Europe although its profile, including once-daily dosing, elevated risk of interactions, and high efficacy makes it an almost perfect monotherapy ASM. The efficacy of CNB would certainly improve under monotherapy conditions due to its superior tolerability.

5.4 | Comparisons of our experiences with CNB with add-on LCM, PER, and BRV

One meta-analysis of pivotal trials involving recently introduced ASMs also highlighted the efficacy of CNB, despite the methodological disadvantage of comparing multicenter trials across diverse populations and study sites, albeit performed with comparable methodology.[16] The conclusions drawn from these indirect comparisons are therefore significantly affected and may be considered unreliable. This was also the case in the work of other similar studies,[19,20,47] although all these studies consistently indicated the superior retention rate or efficacy of CNB compared with other recently introduced ASMs.

We believe that the methodological advantage of our approach is the homogeneity of the site and the PWE groups although our data certainly do not replace a true comparative trial. However, our data strongly support the meta-analyses mentioned above. Comparing our data with the observational studies we performed in the past with recently introduced ASMs, namely LCM, PER, and BRV at our center with a very similar study design, we found that the retention rate for CNB was significantly higher (see Table 3). Also, both the rates of seizure-free individuals and responders achieving at least a 50% reduction in seizures, as well as retention rates, tended to be higher compared with other recently introduced ASMs. Additionally, tolerability was significantly better than with PER.

With a lower number of previous ASMs and a differing number of concomitant ASMs in some instances, the four groups of PWE were still comparable. The favorable effect of CNB was even more pronounced because the group of PWE in our study reflects a slightly more difficult-to-treat cohort than those with LCM and BRV.

The rather low retention rate of BRV in our center may be attributable in part by the possibility of a re-switch to LEV,[32] and the markedly different pharmacokinetics of BRV compared to other ASMs, especially to PER and CNB. In difficult-to-treat epilepsies with a potentially rather high rate of non-responders, the probability to discontinue an add-on drug is presumably higher, if such a drug reaches the plasma steady state very rapidly (as with BRV) than in drugs requiring a slow titration until the maintenance dose is reached (like with PER and CNB).

However, since the study settings and the circumstances were similar, we still conclude that the high efficacy of CNB that was indicated by the pivotal trials[9,12] and confirmed by other real-world studies[18–20,23–28,35,38] was also obvious in our data.

6 | LIMITATIONS

Twenty-two PWE were lost to follow-up (12.8%). This rather high rate results from the fact that our center has a nationwide catchment area. The PWE who dropped out had been treated as in-patients and could not be appropriately followed because they were treated outside our center and did not return to us regularly. Therefore, we do not know whether they continued to use CNB or not. We cannot exclude that the majority discontinued CNB so that the retention rate might have been lower than that reported. However, if we had considered all PWE lost to follow-up as discontinuers, as was sometimes done in other studies,[48] the retention rate would still have been 71.5%, which is well within the range of other ASMs and the retrospective real-world studies about CNB therapy.[16,17,19,20,23,24,26,30–32,48–53]

7 | CLINICAL RELEVANCE AND FUTURE DIRECTIONS

In line with other reports, our experience indicates that CNB is very efficacious and possibly a game changer in epilepsy therapy. True comparative trials and studies in PWE who are treated with CNB at an earlier stage than in our cohort are urgently needed to investigate this assumption.

AUTHOR CONTRIBUTIONS

BJS: Conceptualization; data curation; formal analysis; funding acquisition; investigation; methodology; project administration; resources; supervision; validation; writing—original draft preparation; writing—review and editing. DG: Investigation; data curation; formal analysis; data acquisition; original draft preparation. TI: Patient acquisition; statistics; review and data editing.

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CONFLICT OF INTEREST STATEMENT

BJS: Advisory and consulting honoraria: Angelini, Jazz/GW Pharmaceuticals, Precisis, Roche Diagnostics, UCB. Speaker's honoraria: Al Jazeera, Angelini, Bial, Desitin, Eisai, Jazz/GW Pharmaceuticals, Medscape, Tabuk, Teva, UCB, Zogenix. Research support: Eisai, European Union, Jannsen-Cilag, Jazz/GW Pharmaceuticals, SK Life Sciences, UCB, Zogenix. DG and TI declare no conflict of interest. We confirm that we have read the journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

PATIENT CONSENT STATEMENT

All participants gave their informed consent.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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